

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2020/2021

JANUARY 2021 TRIMESTER

FINAL ASSESSMENT

**UCCD1133 INTRODUCTION TO COMPUTER ORGANISATION AND
ARCHITECTURE**

TUESDAY, 11 MAY 2021

TIME: 8.30 AM – 11.30 AM (3 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING

Instructions to Candidates:

REMINDER: You are reminded to read and adhere to the Final Assessment Instructions to Candidates that has been made available through the UTAR Portal before the commencement of Final Assessment (FA). The detailed instructions for this FA are as follows:

General

1. This Final Assessment (FA) is an Individual, time restricted assessment which consists of FOUR (4) questions. Each question carries 25 marks.
2. You are required to answer **ALL** questions, and submit the **ANSWER SCRIPT** by **11.30 am, 11 MAY 2021**.
3. During the period of **3 hours** of this FA, the examiner(s) can be reached at
 - (a). Microsoft Teams with join code: **kih68y4**, or
 - (b). Email: **ooijo@utar.edu.my**You may use the above e-platform(s) to check with the examiner(s) if you need any clarification on this FA question paper.
4. You may refer to any books, lecture notes, published materials, online resources etc. when answering the questions. Proper referencing is necessary to avoid plagiarism. The preferred referencing style is Harvard. Candidates are reminded that **Copy-and-Paste, Consultation, Discussion and Sharing of Answers** with others are **STRICTLY PROHIBITED** in this FA.

UCCD1133 INTRODUCTION TO COMPUTER ORGANISATION AND ARCHITECTURE

Answer Script File

5. The answer script **MUST** be in a **PDF file format**, in A4 size format. **Note: Please keep the file size NOT exceeding 30MB per file.**
6. Please check your Index Number generated by the Division of Examinations, Awards, and Scholarships (DEAS). You **MUST** name your answer script file using the following file name for submission:

UCCD1133_FA_[Programme Abbreviation]_[Your Index Number]

For example, if you are from the degree programme IA, and your Index Number is A01234CBIAF, then your answer script should be named as:

UCCD1133_FA_IA_A01234CBIAF.pdf

Answer Script File Submission

7. Your answer script file has to be submitted to the following platform(s) before the due time/date.
 - (a). Submit your answer script into WBLE link :<http://wble-kpr.utar.edu.my/wble-kpr/mod/assignment/view.php?id=562777>
Note: Use your “1UTAR” email account to access the WBLE. You can edit your submission before the link is closed.
 - (b). Send your answer script to the course code email:
UCCD1133.FICT@utar.edu.my
Note: For the subject / title of your email, use the file name of your answer script, i.e.,
UCCD1133_FA_[Programme Abbreviation]_[Your Index Number]
8. Please make sure you submit the correct, complete and final version of your answer script and the same copy of answer script file is to be submitted to the above platform(s). When multiple answer script files are received, the examiner will randomly choose one for marking.

UCCD1133 INTRODUCTION TO COMPUTER ORGANISATION AND ARCHITECTURE

Contents of Answer Script

9. The **first page** of your submission is the **Final Assessment Cover Page**. You **MUST** use the template given and fill in the following information
 - Your Programme (Abbreviation)
 - Your Index Number
 - Your Name
 - Your Student ID
10. The **second page** of your submission is the **Final Assessment Declaration Statement**. You **MUST** use the template given, and digitally sign on the form to indicate the authenticity of your submitted work is without plagiarism.
11. Each question should be answered starting on a new page. It is recommended that the answer to each question is limited to **8 pages** or **2000 words**, whichever is lower.
12. For answer scripts that have text-based answers only, all texts **MUST** be typed and recommended using Times New Roman characters with font size **12**, with proper spacing and alignment, except for the drawings and equations/calculations.
13. For answer scripts that require / contain drawings, equations and calculations with short text descriptions, you can hand-write your answers on a blank paper, and then use the scanner apps in your smartphone (Scanner Document, CamScanner, Adobe Scanner etc.) to take scanned copy, or you can type if necessary and include the scanned copy taken in the Word document before save it as PDF, as part of your submitted answers.
14. Please include a page number on **each and every page** of your answer script. Ensure that each page of answer scripts is in sequence prior to online submission.

WARNING OF PLAGIARISM

15. Answer scripts shall be uploaded to Turnitin to check for the originality of submitted answers. In the case of suspected plagiarism, the evidence will be submitted to the Examination Disciplinary Committee of the University. Disciplinary action shall be taken against any candidate who is found to have plagiarized in the answer submitted. Hence, candidates are reminded to abide by all University Rules and Regulations and any instructions/guidelines relating to examinations/assessments.

UCCD1133 INTRODUCTION TO COMPUTER ORGANISATION AND ARCHITECTURE

Q1. (a) Answer the following questions. Show all your workings.

- (i) Assume N to be the hexadecimal number equals to the last three digits of your student ID. Convert N to base 8. (4 marks)
- (ii) Solve the arithmetic operation $326_8 - 1010\ 1110_{\text{Gray}}$ in binary using 2's complement method and give your final answer in binary. Assuming an 8-bit system. (7 marks)
- (iii) Simplify the following expression using Boolean algebra

$$F(A,B,C) = (A+B)(A'+C)(B+C) + (A+BC) \quad (4 \text{ marks})$$

- (b) Figure 1.1 shows a combinational circuit with three inputs. Answer the following question by referring to this figure.

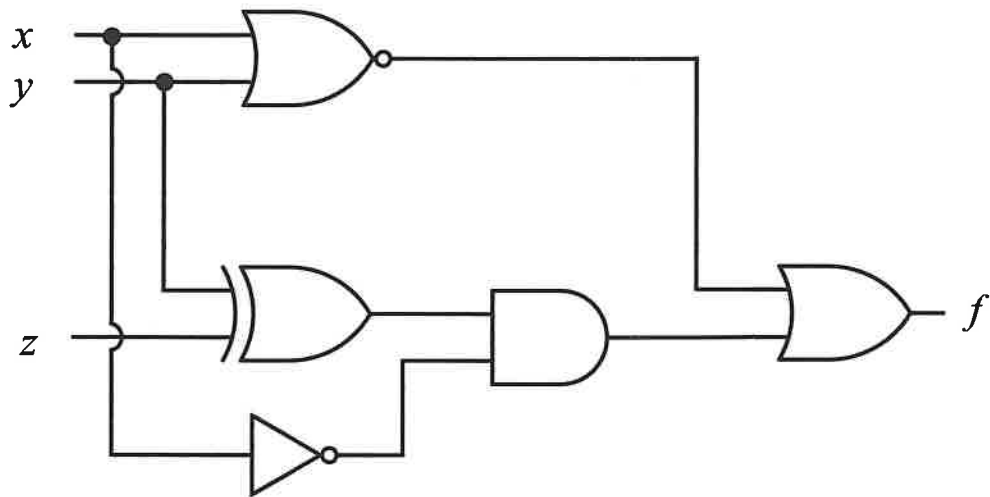


Figure 1.1

- (i) Derive and simplify the Boolean expression for the circuit in Figure 1.1. (4 marks)
- (ii) Construct the corresponding truth table for the circuit in Figure 1.1. (4 marks)
- (iii) From the truth table, obtain the simplified Boolean expression for complement of f , i.e., f' . (2 marks)

[Total : 25 marks]

UCCD1133 INTRODUCTION TO COMPUTER ORGANISATION AND ARCHITECTURE

- Q2. (a) The following machine codes are extracted from MIPS based processor's assembly language program. Answer the following questions by referring to this code.

0x3C15ABCD
0x36B51133

- (i) Translate the machine codes above to MIPS assembly code. Show all your workings. (10 marks)
 - (ii) The above two instruction can be implemented by a single pseudoinstruction. Identify and briefly elaborate the operation of this pseudoinstruction. (3 marks)
 - (iii) Discuss the pseudoinstruction in (a)(ii) by providing TWO (2) disadvantages. (2 marks)
- (b) (i) List the three types of MIPS datapath implementation. Briefly describe each of them. (6 marks)
- (ii) Elaborate the meanings of "Preserved across a call" for registers. (4 marks)
- [Total : 25 marks]

- Q3. (a) Figure 4.1 shows a fragment of a MIPS assembly code. Answer the following questions by referring to this figure.
- (i) Compare the partition fields of basic instruction format types responsible for *Instruction A*, *B* and *C*, on how processing speed can be affected, by providing TWO (2) justifications. (8 marks)
 - (ii) Discuss how it conform to the MIPS Design Principles by having different partition fields for the instruction format types mentioned in (a)(i). (3 marks)
- (b) Compare and provide TWO (2) similarities and TWO (2) differences for addressing modes applied on *Instruction A* and *Instruction B* as shown in Figure 4.1. (8 marks)
- (c) Discuss THREE (3) disadvantages of using dynamically linked libraries (DLLs). (6 marks)

UCCD1133 INTRODUCTION TO COMPUTER ORGANISATION AND ARCHITECTURE

Q3. (Continued)

Address	Assembly code
...	...
0x004000A4	takeNo: li \$v0, 5
0x004000A8	syscall
0x004000AC	move \$t5, \$v0
0x004000B0	lw \$t0, biggestNo
0x004000B4	lw \$t1, smallestNo # Instr. A
0x004000B8	slt \$t2, \$t0, \$t5
0x004000BC	bne \$t2, \$0, alert
0x004000C0	slt \$t2, \$t5, \$t1 # Instr. B
0x004000C4	bne \$t2, \$0, alert
0x004000C8	jal showConclusion
0x004000CC	alert:
0x004000D0	li \$v0, 4
0x004000D4	la \$a0, alertMessage
0x004000D8	syscall
0x004000DC	j takeNo # Instr. C
...	...

Figure 4.1

[Total : 25 marks]

- Q4. (a) In a MIPS based processor's memory hierarchy system, memory allocation is divided into multiple segments.
- Define and describe the text segment. Give the address range and calculate total capacity to hold instructions. (8 marks)
 - Describe how to locate and thus retrieve instruction in text segment. (3 marks)
- (b) In the processor's internal data communication medium, bus arbitration is commonly used.
- Elaborate why the need for bus arbitration mechanism. (2 marks)
 - Describe the two modes for the bus arbitration. (6 marks)
- (c) Describe TWO (2) of the data transfer modes between I/O peripherals and main memory. (6 marks)

[Total : 25 marks]